

Mame Diarra Dieng

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EDUCATION

École Supérieure Polytechnique de Dakar (ESP)

Engineering Degree in Artificial Intelligence & Big Data

Courses: Machine Learning, Distributed Systems, Virtualization Infrastructure, Probability & Statistics, Operating Systems

2024 – Present

Dakar, Senegal

École Supérieure Polytechnique de Dakar (ESP)

University Diploma in Computer Science

2022 – 2024

Dakar, Senegal

PROFESSIONAL EXPERIENCE

SONATEL (Orange Senegal)

Data Engineering Intern

May 2024 – August 2024

Dakar, Senegal

- Designed and deployed a pipeline processing 1M+ records/day from multiple sources using the Elastic Stack (Elasticsearch, Logstash, Kibana)
- Advanced Logstash configurations reducing manual preprocessing by 40%
- Built Kibana dashboards for real-time KPI monitoring, cutting analysis time by 30%
- Complete pipeline documentation for internal knowledge transfer

PROJECTS

Book Recommendation Engine

March 2026

Python, Flask, Pandas, TF-IDF, Sentence Transformers, SQLite

- Collected and cleaned a dataset of 50,000+ books with Pandas (normalization, missing-value handling)
- Semantic modeling via TF-IDF and Sentence Transformers to capture content similarity between books
- Hybrid engine combining content-based filtering with user interactions (reading history, ratings)
- Flask REST API exposing recommendations, SQLite persistence, HTML/CSS/JavaScript web interface

AgriGrenier — IoT Agricultural Monitoring Platform

2024

Flask, Arduino, Supabase, Twilio, n8n | github.com/diarraaaa/AgrigrenierIOT

- Arduino sensors measuring temperature, humidity and storage conditions in real time
- Flask REST API for receiving and processing sensor data, with a Supabase database
- Automated SMS alerts via Twilio triggered as soon as a critical threshold is reached; orchestration via n8n
- Finalist — Polytechnic Alumni Forum · New Deal Technologique

Budget Scolaire — School Budget Management System

2024

Flask, MySQL, Supabase

- Web application for school budget tracking with real-time interactive dashboards
- Secure authentication and role-based access control (administrator, manager, consultant)

TECHNICAL SKILLS

Languages: Python, PHP, C, SQL, HTML/CSS

Frameworks & Backend: Flask, Django, React JS, REST API

Data Science & AI: Pandas, NumPy, Scikit-learn, Sentence Transformers, Matplotlib, SciPy, Ollama / LLM

Big Data & Engineering: Hadoop, ELK Stack (Elasticsearch, Logstash, Kibana), Parallel Computing

Databases: MySQL, PostgreSQL, SQLite, Supabase, Elasticsearch

DevOps & Cloud: Docker, Proxmox, AWS, Linux, Git / GitHub

IoT & Automation: Arduino, Twilio API, n8n

Languages spoken: French (native), English (technical), Arabic (reading), Korean (reading)